

# **B.Sc. (Hons.) Semester III (NEP) Mid-Semester Examination 2025–26**

**DST-Centre for Interdisciplinary Mathematical Sciences (CIMS)**

**Paper Code: CIMSE31 : Python Programming for Data Analytics**

**Time: 1 Hour**

**Max. Marks: 20**

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**Note:** Attempt any FOUR questions. Each question carries equal marks (5 Marks each).

1. What do you understand by Object-Oriented Programming (OOP)? Explain the core features of Python that make it unique and beneficial compared to other programming languages. [5]
2. Explain the key differences among Tuple, List, and Dictionary in Python with suitable code examples. Highlight their mutability and primary use cases. [5]
3. Write a Python script to perform basic data manipulation using Pandas. Discuss the creation of Pandas Series and DataFrame objects. [5]
4. Explain conditional statements and loop control mechanisms (`for`, `while`, `break`, `continue`) in Python with appropriate examples. [5]
5. Discuss the role of Python libraries such as NumPy and Matplotlib in Data Analytics with relevant snippets. [5]